

Hermes Agent Architecture

Public-safe architecture snapshot generated at 2026-08-28T06:15:29-04:00.

Source of truth: local Hermes configuration and runtime status on the operator Linux host.

Sensitive values, private chat IDs, OAuth headers, personal finance/client workflow names, and tokens are intentionally omitted or grouped.

Executive Summary

Hermes Agent runs as a **Linux-primary, multi-platform AI operations hub**. The gateway is a systemd service that connects Telegram, Mattermost, WhatsApp, API Server, Home Assistant, MCP servers, cron jobs, skills, memory/context stores, and remote Windows operations.

The current architecture adds a **durable Kanban orchestration layer**: `default` remains the interactive/orchestrator profile, while `researcher`, `worker`, and `reviewer` are launched as ephemeral Kanban worker-runs for durable research, implementation, and verification tasks. Safe post-session automation generates local review artifacts (session summaries, skill-mining candidates, trace-derived code graphs) without silently installing skills or uploading traces.

The default model remains `openai-codex / gpt-5.5`. Local/experimental providers such as **LM Studio** and **Free Kimi** are configured as optional providers only; they are not the default route for Telegram or production workflows.

High-Level Architecture

[!Hermes Architecture](#)

Runtime Topology

Layer	Components	Notes
User channels	Telegram, Mattermost, WhatsApp, API Server	Gateway adapters route inbound messages into the same Hermes agent core.
Agent core	Hermes Agent gateway + CLI + tool loop	Runs on Linux under <code>hermes-gateway.service</code> .
Models	OpenAI Codex primary, Copilot fallback, optional LM Studio, optional Free Kimi proxy	Provider selection is explicit; experimental endpoints are not default.
Tools	Terminal, files, browser, web, vision, TTS, cron, delegation, Home Assistant, GBrain, NotebookLM, CodeGraph	Enabled toolsets differ by platform/session but the gateway exposes the main operational surface.
Knowledge	GBrain, NotebookLM, session search, imported Claude context, skills	Long-term memory and research context are split across structured stores and markdown skills.
Automation	Hermes cron scheduler, no-agent scripts, hooks, plugins, post-session automation	Backups, monitoring, GitHub publication, finance snapshots, smart-home logging, news/media digests, watchdogs, session summaries, skill-mining reviews, and trace graphs.
Durable orchestration	Kanban dispatcher + <code>default/researcher/worker/reviewer</code> profiles	The gateway dispatcher claims ready tasks and launches profile-specific worker-runs; <code>stopped</code> profiles are normal idle state.
Operational surfaces	Tasks, skills, rules, hooks, plugins, MCP servers, toolsets	Documented as public-safe counts/categories plus selected non-sensitive examples.
Remote systems	Windows workstation over SSH, Home Assistant, Google Drive/GitHub, local LM Studio	Linux remains the control plane; Windows is operated remotely when needed.

Operational Surface Inventory

Surface	Detected public-safe state	Notes
Scheduled tasks / cron	78 jobs; 39 no-agent script jobs; 0 agent-backed jobs	Exact private task names are grouped by category.
Skills	398 detected skill files across 27 categories	Private/client-sensitive skill names are omitted from examples.
Hooks / webhooks	shell allowlist present: False; allowlist entries: 0; plugin hook manifests: 25	Hook command bodies are not published.
Plugins	58 visible plugin rows captured; enabled estimate 5	Descriptions omitted to avoid leaking credential/env surfaces.
MCP servers	11 configured MCP servers	GBrain, NotebookLM, CodeGraph are the active core MCP surfaces.

Scheduled tasks / cron categories

Category	Active jobs	Public-safe purpose
Backup & sync	5	Protect configuration, repositories, databases, and knowledge stores.
GitHub & publishing	6	Maintain GitHub/publication surfaces and repo health digests.
Home automation	2	Log smart-home/home-environment telemetry.
Knowledge & memory	7	Keep GBrain/memory/context stores healthy and up to date.
Media/news monitoring	2	News, RSS, YouTube, and briefing pipelines.
Other scheduled automation	39	Other local automation jobs.
Private finance automation	5	Private finance workflow snapshots; details omitted from public docs.
Reliability watchdogs	12	Auto-healing, environment guards, timeout/watchdog checks.

Skills surface

Hermes currently has a broad skill surface. The public inventory lists category counts and non-sensitive examples only.

Skill category	Count
.archive	11
android	2
apple	5
autonomous-ai-agents	20
creative	40
data-science	1
devops	28
ecc-imports	4
email	5
external	7
gaming	2
github	16
mcp	2
media	7
mlops	23
note-taking	4
operations	3
personal	9
productivity	49
red-teaming	1
research	35
security	4
smart-home	5
social-media	2
software-development	93
uncategorized	19
web-development	1

Public-safe skill examples:

Skill	Category	Public-safe description
dspy	mlops	DSPy: declarative LM programs, auto-optimize prompts, RAG.
fine-tuning-with-trl	mlops	TRL: SFT, DPO, PPO, GRPO, reward modeling for LLM RLHF.
axolotl	mlops	Axolotl: YAML LLM fine-tuning (LoRA, DPO, GRPO).
unsloth	mlops	Unsloth: 2-5x faster LoRA/QLoRA fine-tuning, less VRAM.
instructor	mlops	Extract structured data from LLM responses with Pydantic validation, retry failed extractions automatically, parse complex JSON with type sa
nemo-curator	mlops	GPU-accelerated data curation for LLM training. Supports text/image/video/audio. Features fuzzy deduplication (16× faster), quality filterin
huggingface-hub	mlops	HuggingFace hf CLI: search/download/upload models, datasets.
segment-anything-model	mlops	SAM: zero-shot image segmentation via points, boxes, masks.
chroma	mlops	Open-source embedding database for AI applications. Store embeddings and metadata, perform vector and full-text search, filter by metadata.
faiss	mlops	Facebook
[REDACTED]	mlops	Evidence-first evaluation of local LLM recommendations against the user
evaluating-llms-harness	mlops	lm-eval-harness: benchmark LLMs (MMLU, GSM8K, etc.).
weights-and-biases	mlops	W&B: log ML experiments, sweeps, model registry, dashboards.
llama-cpp	mlops	llama.cpp local GGUF inference + HF Hub model discovery.
obliteratus	mlops	OBLITERATUS: obliterate LLM refusals (diff-in-means).
outlines	mlops	Outlines: structured JSON/regex/Pydantic LLM generation.
serving-llms-vllm	mlops	vLLM: high-throughput LLM serving, OpenAI API, quantization.
qdrant-vector-search	mlops	High-performance vector similarity search engine for RAG and semantic search. Use when building production RAG systems requiring fast nearest
strategic-reading	uncategorized	Read a book, article, transcript, or case study through the lens of a specific strategic problem you
better-colors	external	OKLCH color space and color usage for web projects. Convert hex/rgb/hsl to oklch, generate palettes, check contrast, handle gamut boundaries

Skill	Category	Public-safe description
better - layout	external	Layout structure for web interfaces, from grouping and alignment to reading order, progressive disclosure, and adaptive breakpoints. Use whe
better - ui	external	Design engineering principles for making interfaces feel polished. Use when building UI components, reviewing frontend code, implementing an
better - interface	external	>-
better - typography	external	Web typography from choosing fonts to spacing, wrapping and accessibility. Use when picking or pairing typefaces, configuring variable fonts
better - accessibility	external	Accessibility engineering for product interfaces, from focus states and keyboard support to ARIA, forms, and screen readers. Use when buildi

Hooks, webhooks, and plugin hook manifests

Hermes has multiple hook-related surfaces: shell-hook allowlists, webhook subscriptions, and imported Claude/Claude plugin hook manifests. The public repo records only surface/count information, not hook command bodies.

Hook manifest surface
.claude/plugins/marketplaces/thedotmack/plugin/hooks/hooks.json
.claude/plugins/marketplaces/thedotmack/cursor-hooks/hooks/hooks.json
.claude/plugins/marketplaces/claude-plugins-official/plugins/learning-output-style/hooks/hooks.json
.claude/plugins/marketplaces/claude-plugins-official/plugins/security-guidance/hooks/hooks.json
.claude/plugins/marketplaces/claude-plugins-official/plugins/hookify/hooks/hooks.json
.claude/plugins/marketplaces/claude-plugins-official/plugins/claude-security/hooks/hooks.json
.claude/plugins/marketplaces/claude-plugins-official/plugins/ralph-loop/hooks/hooks.json
[REDACTED]
.claude/plugins/marketplaces/claude-code-warp/plugins/warp/hooks/hooks.json
.claude/plugins/marketplaces/headroom-marketplace/plugins/headroom-agent-hooks/hooks/hooks.json
.claude/plugins/marketplaces/claude-code-plugins/plugins/learning-output-style/hooks/hooks.json
.claude/plugins/marketplaces/claude-code-plugins/plugins/security-guidance/hooks/hooks.json
.claude/plugins/marketplaces/claude-code-plugins/plugins/hookify/hooks/hooks.json
.claude/plugins/marketplaces/claude-code-plugins/plugins/ralph-wiggum/hooks/hooks.json
[REDACTED]
.claude/plugins/marketplaces/ponytail/hooks/hooks.json
.claude/plugins/cache/thedotmack/claude-mem/13.12.4/hooks/hooks.json
.claude/plugins/cache/thedotmack/claude-mem/13.11.0/hooks/hooks.json
.claude/plugins/cache/thedotmack/claude-mem/13.12.2/hooks/hooks.json
.claude/plugins/cache/thedotmack/claude-mem/13.13.0/hooks/hooks.json

Plugin surface

Plugin	Status
browser-browser-use	not enabled
browser-browserbase	not enabled
browser-firecrawl	not enabled
chronos	not enabled
basic	not enabled
drain	not enabled
nous	not enabled
self-hosted	not enabled
disk-cleanup	not enabled
google_meet	not enabled
deepinfra	not enabled
fal	not enabled
krea	not enabled
openai	not enabled
openai-codex	not enabled
openrouter	not enabled
xai	not enabled
langfuse	not enabled
a2a-platform	not enabled
buzz-platform	not enabled
dingtalk-platform	not enabled
discord-platform	not enabled
email-platform	not enabled
feishu-platform	not enabled
google_chat-platform	not enabled

Plugin	Status
irc-platform	not enabled
line-platform	not enabled
matrix-platform	not enabled
mattermost-platform	not enabled
ntfy-platform	not enabled

Low-Level Surface Files

The repository includes dedicated, low-level public-safe files for each operational surface:

File	Contents
docs/surfaces/tasks.md	Scheduled tasks/cron jobs, modes, scripts, schedules, delivery class, workdir class.
docs/surfaces/skills.md	Skill categories, counts, public-safe examples, operational semantics.
docs/surfaces/hooks.md	Shell hook allowlist state, webhook state, plugin hook manifests.
docs/surfaces/plugins.md	Hermes plugin registry rows and status.
docs/surfaces/mcp-and-toolsets.md	MCP servers and toolset count estimates.
docs/surfaces/models-and-gateway.md	Model routing, gateway status, channel/platform surface.
docs/surfaces/orchestration.md	Kanban profiles, rules layer, self-improvement automation, and operating contract.

Model Routing

Role	Provider	Model	Notes
Primary	nous	stealth/ox-alpha	Default for Telegram/API/CLI gateway sessions
Fallback			Used when primary fails
Optional provider	lmstudio	qwenvl3bunc	http://127.0.0.1:1234/v1
Optional provider	nvidia	meta/llama-3.3-70b-instruct	https://integrate.api.nvidia.com/v1
Optional provider	freekimi	cfbt-kimi	http://127.0.0.1:3271/v1
Optional provider	forge_freekimi	cfbt-kimi	http://127.0.0.1:8081/v1
Optional provider	forge_lmstudio	qwenvl3bunc	http://127.0.0.1:8082/v1
Optional provider	chatgpt_web	chatgpt-5.6-sol-high-web	https://codex.guber.dev/v1
Optional provider	codex_web_gpt	chatgpt-web/light	http://100.82.137.49:17842/v1

Local model trial status

Item	Status
LM Studio endpoint	available at http://127.0.0.1:1234/v1
Reported model IDs	gemma4unc, [REDACTED]
Chat smoke test	`blocked_or_unavailable: {

"error": { "message": "Failed to load model \"gemma4unc\". Error: Model loading was stopped due to insufficient system resources. Under the current settings, this model requires approximately 14.36 GB of memory, and continuing | | Safety decision | Main Hermes remains openai-codex/gpt-5.5`; local provider is optional until a model can load reliably. |

MCP and External Tooling

MCP server	Transport	Public-safe purpose
gbrain	Local HTTP MCP on 127.0.0.1:3131/mcp	Personal knowledge graph, memory, code graph, facts, schema tools.
notebooklm	npx notebooklm-mcp@latest	Grounded research over registered Google NotebookLM notebooks.
codegraph	Local Node command	Code intelligence over selected repositories.

Scheduled Automation

Category	Active jobs	Public-safe purpose
Backup & sync	5	Protect configuration, repositories, databases, and knowledge stores.
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Home automation	2	Log smart-home/home-environment telemetry.
Knowledge & memory	7	Keep GBrain/memory/context stores healthy and up to date.
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Agentic Operating Model

Item	Value	Meaning
Interactive orchestrator	default	Main Telegram/API/CLI profile; decides whether to answer directly, delegate, create Kanban work, or schedule cron.
Research lane	researcher	Evidence-backed OSS/docs/web due diligence and adoption recommendations.
Execution lane	worker	File/terminal/code/config implementation and verification for durable work items.
Review lane	reviewer	Independent review, regression checks, security/config sanity, and final validation.
Default assignee	worker	Fallback when a task/decomposer does not choose a specialist.
Failure limit	5	Raised above the previous aggressive default to avoid noisy retry failures.
Per-profile concurrency	1	Keeps one profile from saturating model/API/browser capacity.
Post-session automation	enabled	Generates review artifacts only; does not auto-install remote skills or mutate active prompts.

The important runtime distinction is that `researcher`, `worker`, and `reviewer` do **not** need to be continuously running gateway profiles. They are idle until a Kanban task is assigned to them; then the gateway dispatcher starts a worker-run for that profile and the worker completes, blocks, or retries the task.

Current Profiles

The live system currently exposes the public-safe profile roster as:

Profile	Model	Gateway	Alias	Distribution
◆default	stealth/ox-alpha	running	–	–
claude	–	stopped	–	–
coding	gpt-5.6-sol	stopped	coding	–
ghidra-restricted	gpt-5.6-sol	stopped	–	–
researcher	gpt-5.6-terra	stopped	hermes-researcher	–
reviewer	gpt-5.6-sol	stopped	hermes-reviewer	–
security-restricted	gpt-5.6-sol	stopped	hermes-security	–
worker	gpt-5.6-terra	stopped	hermes-worker	–

Current profile contract:

Profile	Role	Runtime behavior
default	Interactive orchestrator	Stays running in Telegram/API/CLI gateway; routes work.
researcher	Research/due-diligence worker	Normally stopped; launched by Kanban for evidence gathering.
worker	Implementation/execution worker	Normally stopped; fallback assignee for durable tasks.
reviewer	Independent verification worker	Normally stopped; launched for review/regression/security checks.

Reliability and Safety Boundaries

- Public repo does **not** publish secrets, chat IDs, API keys, OAuth tokens, cookies, private finance details, or client-specific payloads.
- The gateway is supervised by systemd and multiple watchdog cron jobs.
- GBrain runs through a shared local HTTP MCP server to avoid PGLite lock contention.
- Backups and sync jobs run through cron with Google Drive/GitHub targets.
- Windows is treated as a remote worker controlled via SSH, not as the primary Hermes control plane.
- Experimental providers are explicitly optional and must pass direct API + Hermes smoke tests before being used for real agent work.

Operational Notes

- Hermes version/status summary:

Hermes Agent v0.20.5 (2026.8.19)
Install directory: ~/.hermes/hermes-agent
Install method: git
Python: 3.11.15
OpenAI SDK: 2.24.0

- Fallback chain:

Primary: stealth/ox-alpha (via nous)

Fallback chain (3 entries):

1. ox-alpha-free (via opencode-go)
2. gpt-5.6-terra (via openai-codex)
3. gpt-5.3-codex-spark (via openai-codex)

Tried in order when the primary fails (rate-limit, 5xx, connection errors).

Docs: <https://hermes-agent.nousresearch.com/docs/user-guide/features/fallback-providers>

- MCP list:

MCP Servers:

Name	Transport	Tools	Status
keep-mcp	~/.hermes/scripts/...	16 selected	enabled
codegraph	~/.nvm/versions/no...	all	enabled
gbrain	http://127.0.0.1:3131/mcp	all	enabled
notebooklm	npx -y notebooklm-mcp@latest	all	enabled
windows-cua	~/.local/bin/windo...	all	enabled
monarch	https://api.monarch.com/mcp	17 selected	enabled
cloudflare-api	https://mcp.cloudflare.co...	all	enabled
pixelrag	~/github/pixelrag-...	all	enabled
vibe_trading	~/.hermes/scripts/...	29 selected	enabled
display	https://api.display.dev/v...	all	enabled
open_notebook	~/.hermes/oss-eval...	all	enabled

Maintenance

This repository is regenerated by `scripts/update_architecture.py`. The updater is designed to be safe for a public repository:

1. collect local Hermes runtime state,
2. redact secrets and private identifiers,
3. group sensitive scheduled jobs by category,
4. regenerate Markdown, SVG, HTML, JSON inventory, and PDF,
5. commit and push only if files changed.

Source Files

- `ARCHITECTURE.md` - canonical public architecture document.
- `docs/hermes-architecture.svg` - standalone diagram.
- `docs/hermes-architecture.html` - browser-friendly diagram page.
- `docs/Hermes-Architecture.pdf` - rendered PDF copy.
- `docs/surfaces/*.md` - low-level public-safe inventories for tasks, skills, hooks, plugins, MCP/toolsets, and gateway/model routing.
- `data/inventory.public.json` - redacted machine-readable snapshot.
- `scripts/update_architecture.py` - public-safe regeneration script.

Rendered from GBrain markdown via Hermes brain-pdf.